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Data Analysis and Visualization

▼ Relevant source files

`hclient/widgets/entity/configEntity.js``hclient/widgets/record/recordDataTable.html``hclient/widgets/record/recordDataTable.js``hclient/widgets/viewers/mediaViewer.js``hclient/widgets/viewers/miradorViewer.php``hclient/widgets/viewers/resultListDataTable.js``hserv/utilities/geo/UTMtoLL.php``import/bookmarklet/bookmarklet-popup.css``viewers/crosstab/crosstabs.css``viewers/crosstab/crosstabs.js``viewers/crosstab/crosstabs.php``viewers/crosstab/crosstabsController.php`

This document describes the data analysis and visualization capabilities in Heurist, focusing on the crosstab analysis system that enables users to perform tabular data analysis and create visual representations of record relationships. For information about geographical mapping features, see [Mapping System](#), and for website generation from database content, see [CMS Website Generator](#).

Overview

The Data Analysis and Visualization system provides tools for analyzing relationships between record fields through a tabular cross-tabulation interface with support for data grouping, statistical calculations, and visual representations. The system allows users to:

- Select fields for row, column, and page dimensions
- Configure custom intervals for numeric, date, and categorical data

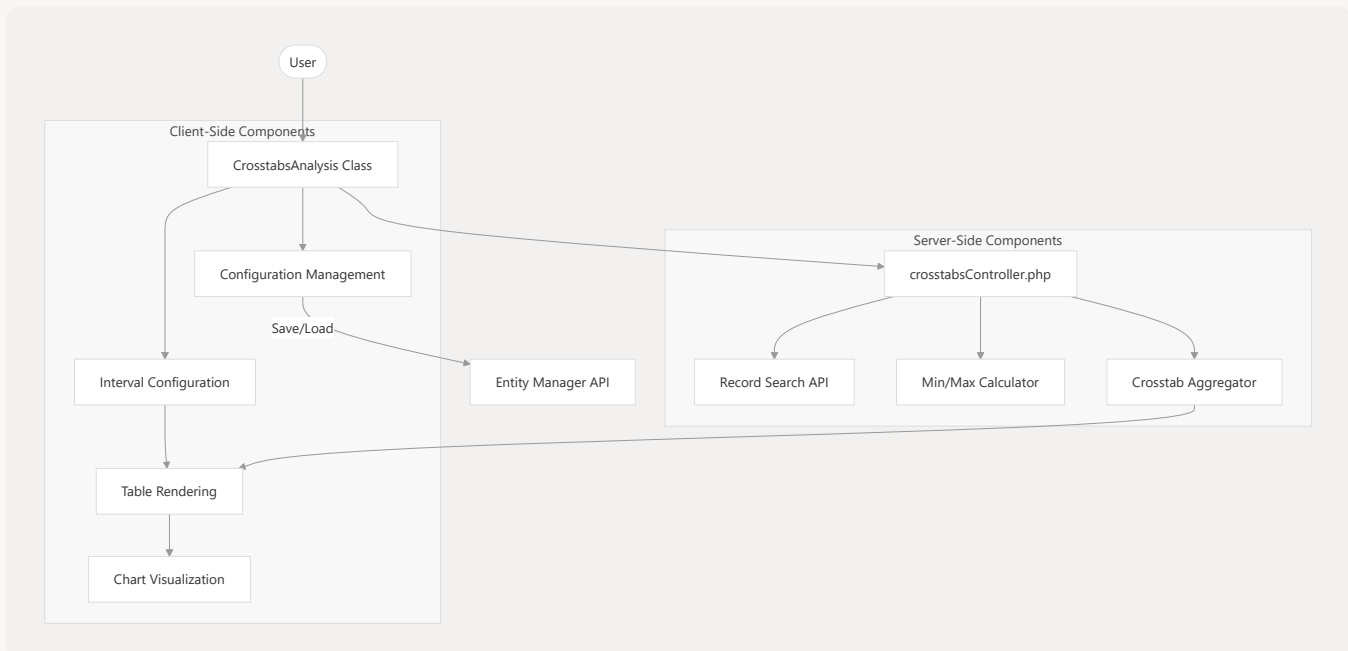
- Perform count, sum, or average calculations
- Visualize results as tables or charts
- Save and load analysis configurations
- Export results in various formats

Sources: `viewers/crosstab/crosstabs.js` 14-58 `viewers/crosstab/crosstabs.php` 14-27

System Architecture

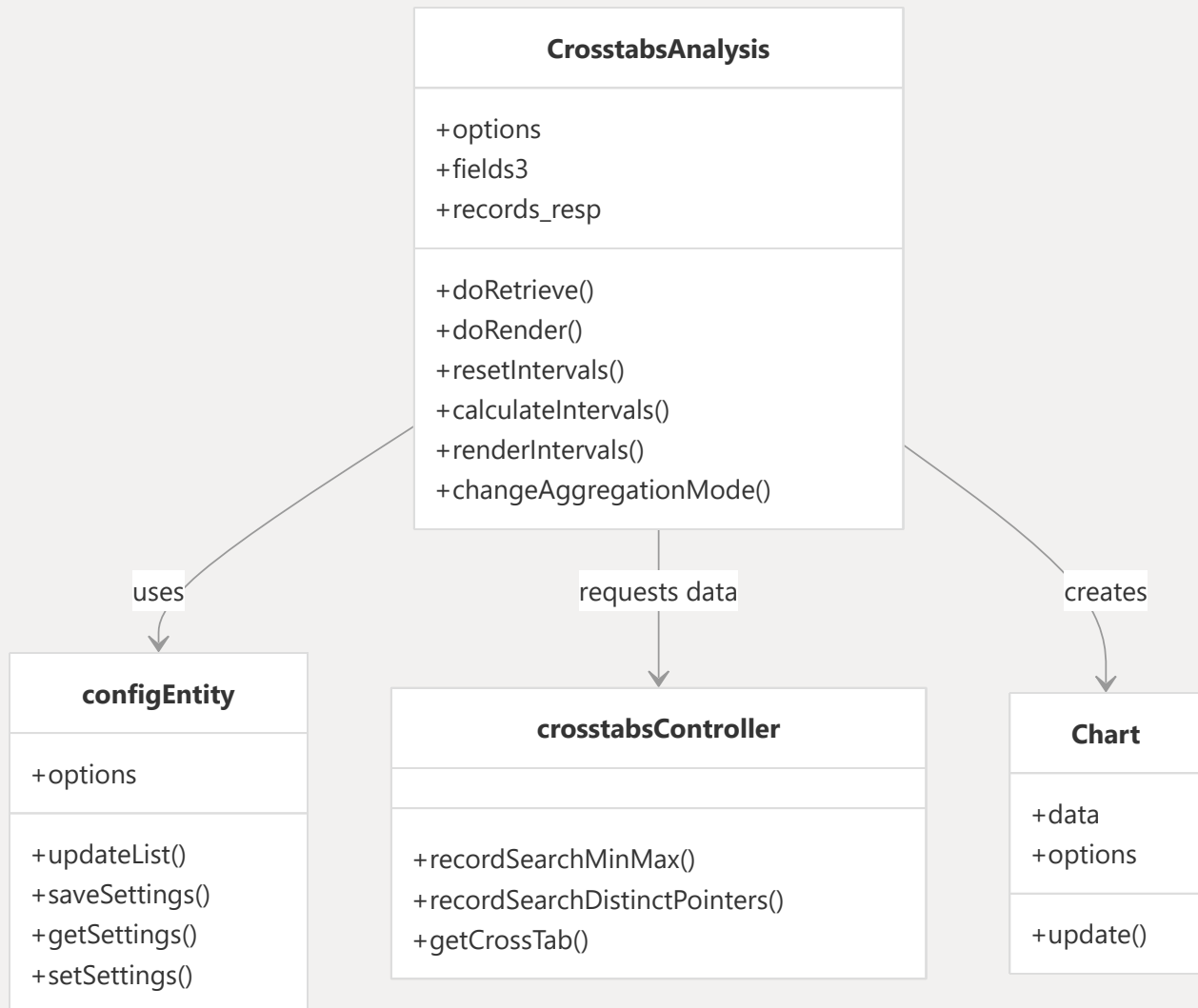
The crosstab analysis system consists of client-side JavaScript components that handle the UI and data presentation, coupled with server-side PHP processing for data retrieval and aggregation.

Architecture Overview



Sources: `viewers/crosstab/crosstabs.js` 39-49 `viewers/crosstab/crosstabsController.php` 10-55

Component Relationships



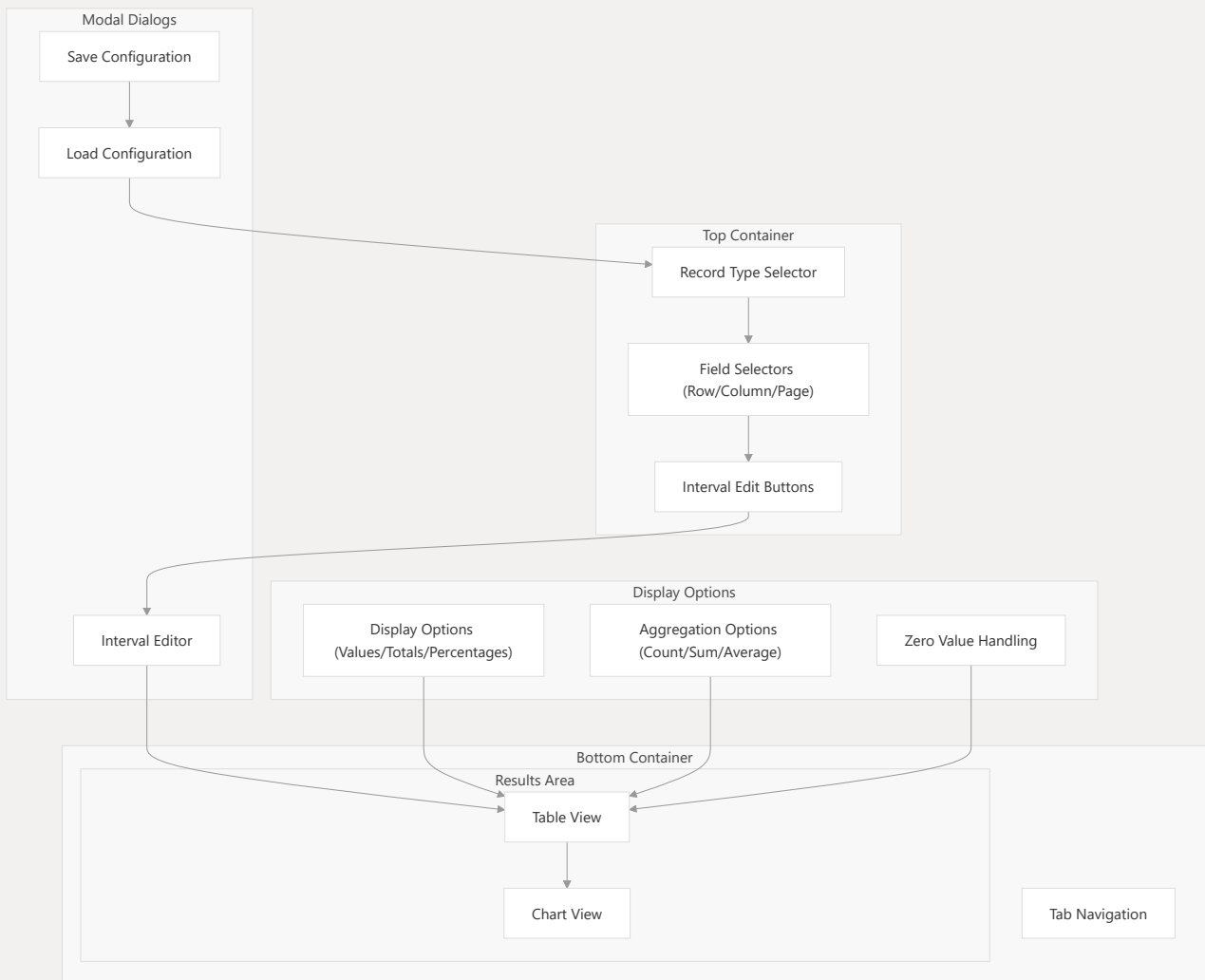
Sources: `viewers/crosstab/crosstabs.js` 39-79 `hclient/widgets/entity/configEntity.js` 22-35

User Interface Components

The crosstab analysis interface is divided into several functional areas:

1. **Configuration Area** - Selection of record types and fields
2. **Interval Definition** - Setting up data grouping
3. **Display Options** - Configuring how results are shown
4. **Results Area** - Tabular and chart visualization

UI Layout



Sources: `viewers/crosstab/crosstabs.php` 84-322 `viewers/crosstab/crosstabs.js` 66-136

Field Selection Interface

The main interface allows users to select record types and up to three variables for analysis:

1. **Record Type Selector** - Determines available fields
2. **Row Variable (Var 1)** - Primary grouping dimension
3. **Column Variable (Var 2)** - Secondary grouping dimension
4. **Page Variable (Var 3)** - Optional tertiary grouping dimension

Each variable can be configured with custom intervals to group data points.

Sources: `viewers/crosstab/crosstabs.php` 190-276

Display Configuration

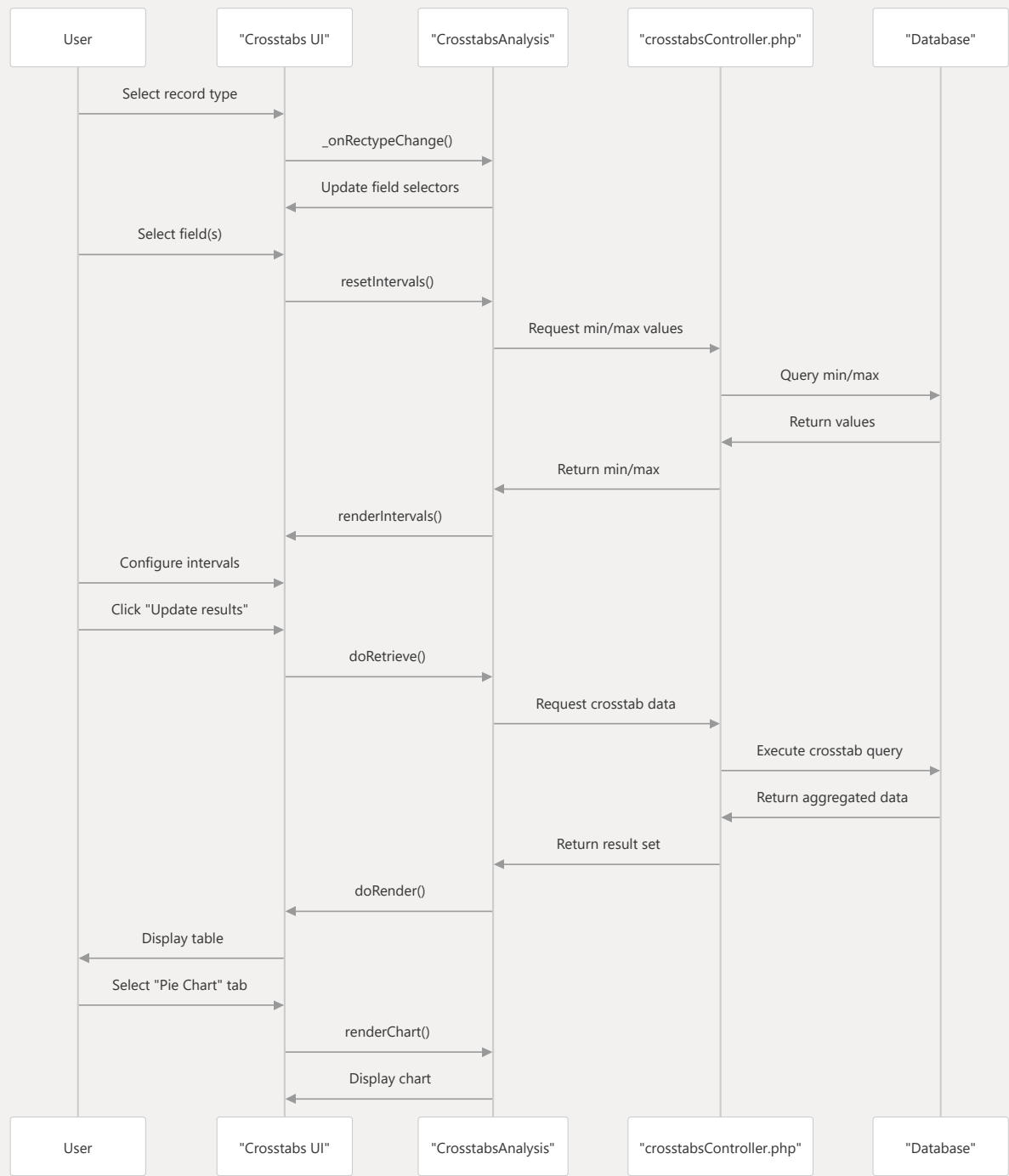
Users can customize how data is displayed:

Option	Description
Values	Show raw count/sum/average values
Totals	Show row and column totals
Row %	Show values as percentage of row total
Column %	Show values as percentage of column total
Blank for zero values	Hide zero values in the table
Show blank rows/columns	Include rows/columns with no data

Sources: `viewers/crosstab/crosstabs.php` 110-184

Data Processing Flow

The crosstab analysis follows a specific flow for retrieving, processing, and visualizing data:



Sources: `viewers/crosstab/crosstabs.js` 126-191
`viewers/crosstab/crosstabsController.php` 179-303

Interval Configuration

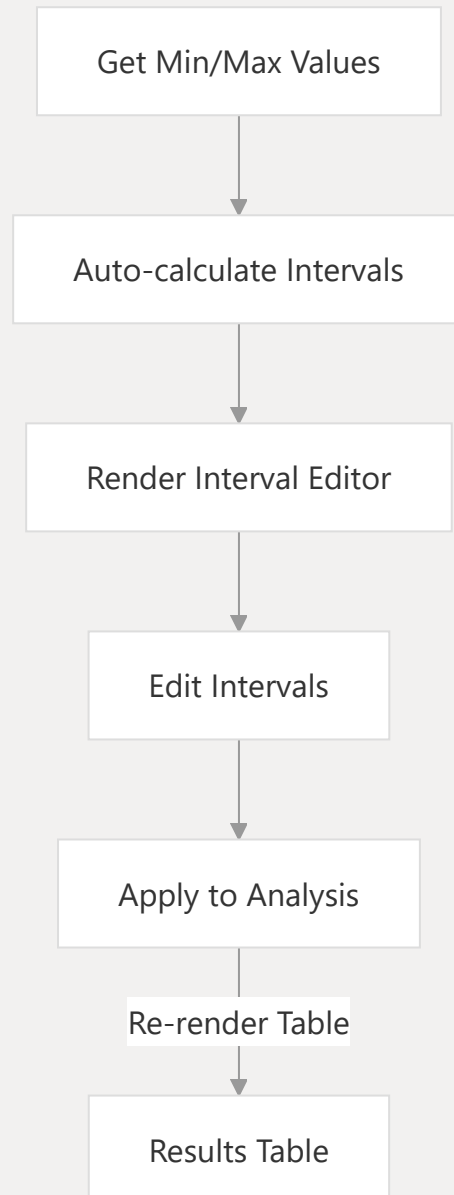
A key feature of the crosstab system is the ability to define custom intervals for grouping data. This is handled differently based on the field type:

Numeric and Date Intervals

For numeric and date fields, the system automatically calculates intervals based on the range of values:

1. Retrieves minimum and maximum values from the database
2. Divides the range into equal intervals (default: 10)
3. Allows manual adjustment of interval boundaries

Users can add, edit, or remove intervals to create custom groupings.



Sources: `viewers/crosstab/crosstabs.js` 360-480 `viewers/crosstab/crosstabs.js` 584-640

Categorical Intervals

For enumeration fields (terms) and record pointers, the system:

- 1. Retrieves all possible values from the vocabulary or related records
- 2. Creates an interval for each unique value
- 3. Allows grouping multiple values into custom intervals

This enables analysis of categorical data by merging related categories.

Sources:

viewers/crosstab/crosstabs.js396-435

viewers/crosstab/crosstabsController.php123-175

Data Aggregation Methods

The system supports three methods of data aggregation:

Method	Description	Field Types
Count	Counts the number of records in each cell	All field types
Sum	Calculates the sum of a numeric field	Integer, Float
Average	Calculates the average of a numeric field	Integer, Float

When sum or average is selected, an additional field must be chosen to aggregate.

Sources:

viewers/crosstab/crosstabs.js143-174

viewers/crosstab/crosstabsController.php189-210

Visualization Options

Tabular View

The primary visualization is a cross-tabulation table showing the relationship between selected variables:

- Rows correspond to the first variable
- Columns correspond to the second variable
- Pages (tabs) correspond to the third variable (if selected)
- Cells contain the aggregated values

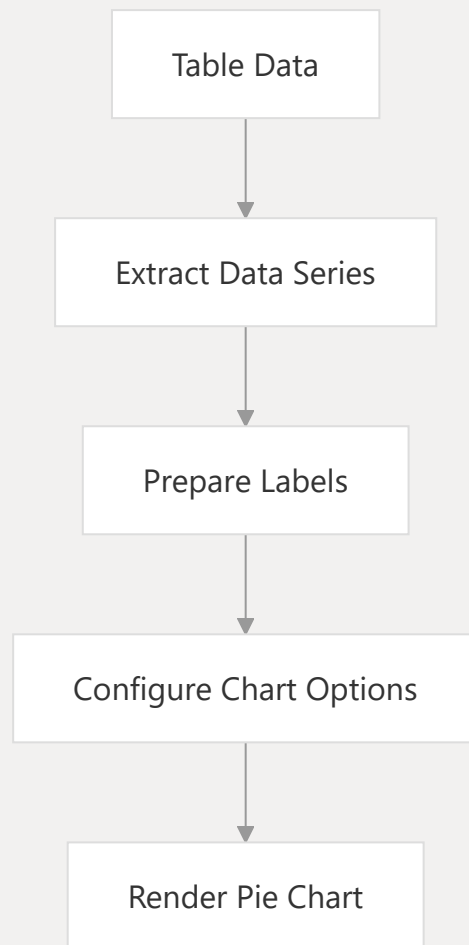
The table includes row and column totals and can display percentages.

Sources:

viewers/crosstab/crosstabs.js2180-2435

Chart Visualization

The system integrates with Chart.js to provide pie chart visualization of the analysis results:

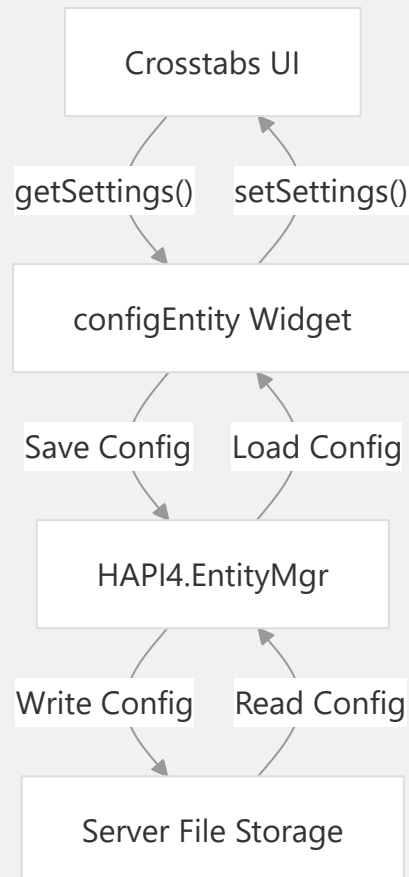


The pie chart visualization is particularly useful for showing the distribution of values for a single variable (row selection).

Sources: `viewers/crosstab/crosstabs.js` 2437-2490 `viewers/crosstab/crosstabs.php` 35-36

Configuration Management

The crosstab analysis system integrates with Heurist's configuration management system to save and load analysis settings:



Configurations are stored as JSON files in the database file storage system, organized by entity type and record type.

Sources: `viewers/crosstab/crosstabs.js` 104-117 `hclient/widgets/entity/configEntity.js` 76-117
`hclient/widgets/entity/configEntity.js` 401-467

Configuration Storage Format

A saved configuration includes:

- Record type ID
- Field selections for rows, columns, and pages
- Interval definitions for each field
- Display options
- Aggregation method

This allows users to quickly switch between different analysis views of their data.

Sources: `viewers/crosstab/crosstabs.js` 2492-2540

Data Export Capabilities

The crosstab system integrates with the DataTables library to provide export functionality:

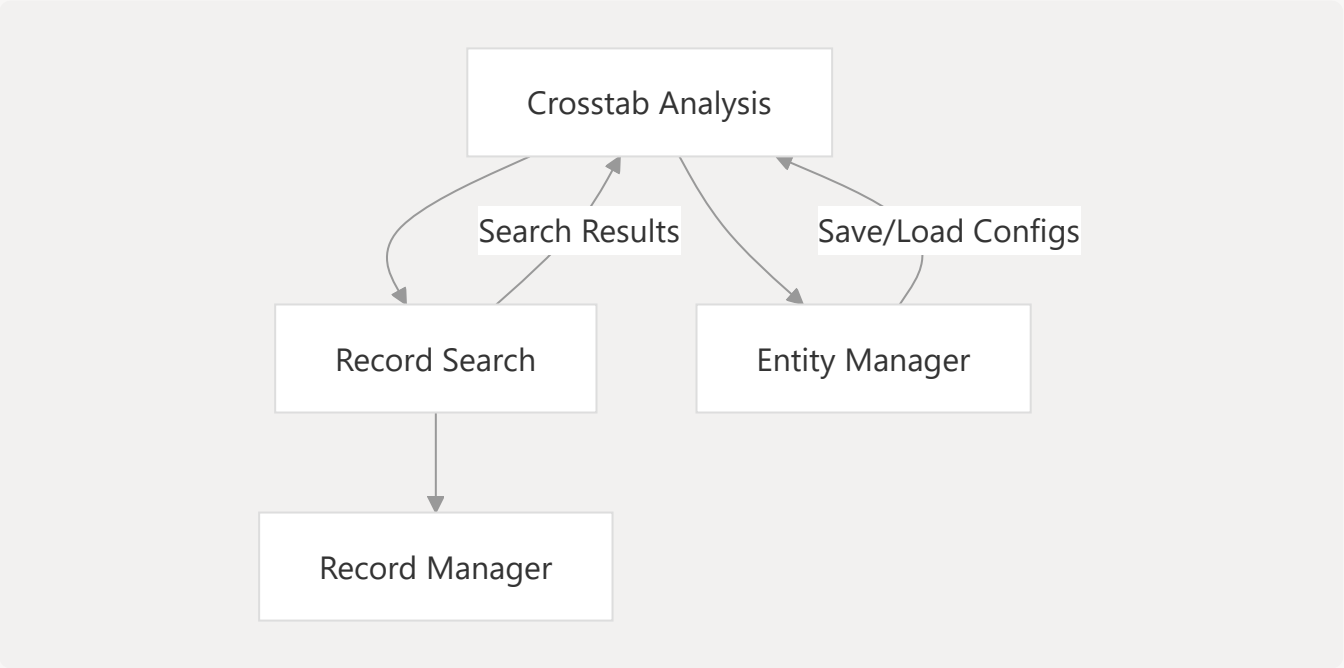
Format	Description
Copy	Copy to clipboard for pasting into other applications
Excel	Export as Excel spreadsheet
PDF	Export as PDF document

The export functionality makes it easy to use analysis results in reports and presentations.

Sources: `viewers/crosstab/crosstabs.php` 41-47

Integration with Other Systems

The crosstab analysis system integrates with other Heurist components:



This integration allows seamless analysis of search results and persistent storage of analysis configurations.

Sources: `viewers/crosstab/crosstabs.js` 58-65

`viewers/crosstab/crosstabsController.php` 223-226